

The edge as a Redheffer step

Last Potapov pivot $\delta = 1 - q^\dagger = -\text{sch}$;
the quarter is not a universal multiple

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August 30, 2026

Abstract

The r362/r369 augmentation $R^\dagger$ is the terminating 2×2 -block Redheffer star of the bulk colligation with the 3×3 edge factor built from the two last dual CD directions and the Sherman–Morrison border channel. The last Schur/Potapov pivot of the closed system is extracted from the border coordinate of $M^\dagger = R^\dagger - \frac{1}{2}I$ (the r362 $\text{sch}_b/q^\dagger$), not by quoting sch . The identification is then $\delta := 1 - q^\dagger = 4\text{sch}_b/(1 + 2\text{sch}_b) = -\text{sch}$ with $c' = 1$. The companion form $\delta = c(\frac{1}{4} - |Z_{\text{loc}}|^2)$ is not a world-blind identity. No RH claim.

Status. Lemma 1 and Lemma 2 are **SATZ** (finite algebra over $\mathbb{Q}$, machine precision on the named windows). Proposition 1 is a named refutation of a universal c . Proposition 2 is a sealed census. Ausgang REDHEFFER_EXACT / PIVOT_IS_ONE_MINUS_QDAG / QUARTER_NOT_UNIVERSAL. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The Redheffer step

Write $A_0 = R_{N-3} - \frac{1}{2}I$ and U_{CD} for the two last dual OP columns on Y (r367). The mixed form of r369 is

$$A = \text{blkdiag}(A_0, -\tfrac{1}{2}), \quad U = \begin{pmatrix} U_{\text{CD}} & s \\ 0 & -1 \end{pmatrix}, \quad E = J = \text{diag}(1, 1, 1/\text{den}),$$

with $s = Rv$ the Sherman–Morrison border vector and $\text{den} = (1 + \gamma) - v^T Rv$. The bulk colligation is the 2×2 block $\Theta = \begin{bmatrix} A & U \\ U^T & 0_3 \end{bmatrix}$. The terminating Redheffer star with load E is

$$\mathcal{R}_E(\Theta) = A + UEU^T. \tag{1}$$

Equivalently, first close the two-dimensional CD port $(A_0 + U_{\text{CD}}U_{\text{CD}}^T)$, pad the $-\frac{1}{2}$ corner, then close the one-dimensional border port with $1/\text{den}$.

Lemma 1 (Redheffer identification). *On the rational 4-node mixed toy, (1) equals the r362 inverse $R^\dagger - \frac{1}{2}I$ with residual 0 ($\text{den} = \frac{23}{30}$). The sequential CD-then-border product equals the packed 3×3 load, residual 0. At the flagship window $k_z = 9$ all three residuals are $1.9 \cdot 10^{-15}$.*

The last 1×1 Schur complement of the 3×3 characteristic matrix $\Phi = J^{-1} + U^T A^{-1} U$ after eliminating the K_2 block is the r401 scalar sch (Haynsworth). That equality is an identification, not a definition of the pivot: the pivot is read from $M^\dagger$ independently.

2 The last pivot

Let sch_b be the Schur complement of $M^\dagger$ onto the border coordinate (r362 (A6)) and

$$q^\dagger = \gamma + v^T(I - E_n)^{-1}v$$

the dual-frame load (r362 $q^\dagger$). Then $\text{sch}_b = (1 - q^\dagger)/(2(1 + q^\dagger))$ identically, so

$$\delta := 1 - q^\dagger = \frac{4 \text{sch}_b}{1 + 2 \text{sch}_b}. \quad (2)$$

This is the energy form $\delta = 1 - |\alpha_{\text{edge}}|^2$ with $|\alpha_{\text{edge}}|^2 = q^\dagger$: the cancellation that a triangle inequality would spend sits in the complex load, not in an absolute majorant.

Lemma 2 (pivot dictionary). *On the rational toy $\text{sch} = -\frac{187}{450}$, $\text{sch}_b = \frac{187}{1426}$, and (2) gives $\delta = \frac{187}{450} = -\text{sch}$ ($c' = 1$). On every named window of the census, $1 - q^\dagger = -\text{sch}$ to f64 residual (flagship $9.5 \cdot 10^{-13}$; selected $k = 9$ $5.5 \cdot 10^{-8}$, graded).*

The constant-vector Woodbury discriminant of r415 is a different number (flagship $\Delta_1 = 2.11 \cdot 10^{-3}$ against $\delta = 0.06696$). Dropping the border column leaves no 3×3 last pivot ($\text{ind}_- K_2 = \text{ind}_- A_0$).

3 The quarter is not a universal multiple

The r263 dictionary is $q_N = \frac{7}{5}Z^2$, so $q_N < 1$ if and only if $|Z| < \sqrt{5/7}$. Since $\frac{1}{4} < \frac{5}{7}$, the bound $|Z_{\text{loc}}| \leq \frac{1}{2}$ (r428/r429, selected census) is a sufficient M -bound on Z_{loc} , not an identity with sch .

Proposition 1 (quarter not universal). *There is no world-blind $c > 0$ with $\delta = c(\frac{1}{4} - |Z_{\text{loc}}|^2)$ on the named census. On selected $k = 3, 4, 5, 6, 7, 9$ both sides are positive and the conversion $c_k = \delta_k/(\frac{1}{4} - |Z_{\text{loc}}, k|^2)$ lies in $[0.155, 5.51]$ ($k = 3$: 5.51; $k = 4$: 0.297). On vacuous MAIN ($k_z = 12, 16$) and living χ_{3-9} , $\delta > 0$ while $\frac{1}{4} - |Z_{\text{loc}}|^2 < 0$. The r263 form $\frac{5}{7} - |Z_{\text{loc}}|^2$ is likewise not a multiple of δ ($q^\dagger \neq q_N$; flagship 0.933 against 0.214).*

4 Worlds

Proposition 2 (two-sided census). *Dead χ (5 of χ_3 plus χ_4 -20): $\delta_k < 0$ on 6/6 ($q^\dagger > 1$). Living MAIN (selected $k = 3..7, 9$ and a six-rung core sample) and living χ_{3-9} : $\delta > 0$. Scramble has $\text{nneg}(A_0) = 21$; the bulk is already dead and the edge pivot of a 3×3 is irrelevant. At the $k = 3$ edge of the selected half-bound, $|Z_{\text{loc}}| = 0.486846$ (air to $\frac{1}{2}$ equal to 0.013154) with $\delta = 0.071564 > 0$.*

5 What this does not close

The Redheffer writing makes the last transformation identifiable as a Schur step, which was the stop condition of DCCXCVII (D). It does not prove $\delta > 0$ cofinally, nor $|Z_{\text{loc}}| \leq \frac{1}{2}$ cofinally, nor $R^\dagger \succeq \frac{1}{2}I$ on the selected sequence. The quarter remains a selected census (r429), not a pivot. Mincut unchanged (base 4 / refined 5). No L^* claim. No $R^\dagger$ claim. No RH claim.